ENEL445 Mock Test — Questions & Solutions

Based on 2024 mid-year exam and 2025 mid-year exam. Questions and mark allocations taken directly from past papers.

Q1. Gradient-Free Optimisation (Part I)

(a) [3 marks] Give three situations where gradient-free methods are more suited than gradient-based methods for solving an optimisation problem.

Solution:

1. **Non-differentiable or discontinuous** objective — gradient does not exist or is unreliable (e.g. simulation outputs, physical experiments, $|x|$ at $x = 0$).
2. **Black-box function** — no analytical expression is available; only function evaluations are possible (e.g. legacy simulation code, hardware-in-the-loop testing).
3. **Noisy function evaluations** — gradient estimates are corrupted by noise, making gradient-based methods unreliable or divergent.

(b) [2 marks] Describe (or sketch) the Nelder-Mead *expansion* operation.

Solution: The worst vertex x_w is reflected through the centroid $\bar{x}$ of the remaining vertices:

$$x_r = \bar{x} + (\bar{x} - x_w) \quad \text{(reflection)}.$$

Expansion then extends further:

$$x_e = \bar{x} + 2(\bar{x} - x_w).$$

If $f(x_e) < f(x_r)$, accept x_e; otherwise accept x_r. On a contour plot x_e lies on the opposite side of $\bar{x}$ from x_w, twice as far as the reflection point.

(c) [3 marks] Name three convergence metrics used with Nelder-Mead.

Solution:

1. **Simplex size**: $\max_i \|x_i - x_1\|_2 < \varepsilon$
2. **Function value spread**: $\max_i |f(x_i) - f(x_1)| < \varepsilon$
3. **Iteration / evaluation limit**: $k > k_{\max}$ or number of function evaluations exceeds $N_{\max}$

(d) [2 marks] Sketch a minimal positive spanning set in $\mathbb{R}^2$ **and** state the process of opportunistic polling in the generalised pattern search algorithm.

Solution: A minimal positive spanning set in $\mathbb{R}^2$ requires $n+1 = 3$ vectors. Example: $\{e_1, e_2, -e_1 - e_2\}$ where $e_1 = (1,0)^\top$, $e_2 = (0,1)^\top$. These span all of $\mathbb{R}^2$ in the positive sense.

Opportunistic polling: poll directions are evaluated one at a time in a preferred order. As soon as one direction yields $f < f_{\text{current}}$, the algorithm immediately accepts that point and starts a new iteration — it does *not* evaluate all remaining directions in the pattern.

(e) [3 marks] State the two significant shortcomings of Shubert's algorithm and briefly explain how DIRECT addresses them.

Solution: Shubert shortcomings:

1. Requires the global Lipschitz constant K to be known in advance — which is generally unknown in practice.
2. Only applicable to *one-dimensional* problems.

DIRECT addresses both:

1. Estimates local Lipschitz constants implicitly per hyper-rectangle by using the f-d plot — no global K is needed.
2. Generalises to n dimensions by partitioning n-dimensional hyper-rectangles.

(f) [3 marks] On the f-d plot, identify which rectangles DIRECT subdivides next when $\varepsilon|f_{\min}| = 3$.

Solution: DIRECT selects all rectangles whose sample point $(d_i, f(c_i))$ lies on the *lower-right convex hull* of the f-d scatter and satisfies:

$$f(c_i) - K d_i \le f_{\min} - \varepsilon|f_{\min}| \quad \text{for some } K > 0.$$

With $\varepsilon|f_{\min}| = 3$, points are eligible only if their value is within 3 of $f_{\min}$ after accounting for the hull slope. On the plot: circle all points on the lower-right convex hull (points not dominated below-right by any other point).

(g) [4 marks] Briefly explain the three main steps in a genetic algorithm and discuss the advantages and disadvantages of a binary-coded gene versus a real-coded gene.

Solution: Three steps:

1. **Selection**: choose parents with probability proportional to fitness (e.g. roulette-wheel or tournament selection).
2. **Crossover**: combine genetic material of two parents at a random cut point to produce offspring (e.g. single-point crossover swaps tails).
3. **Mutation**: randomly flip/alter genes with a small probability to maintain diversity and prevent premature convergence to a local optimum.

Binary vs real-coded:

- **Binary**: simple, well-studied operators; but suffers from the *Hamming cliff* — numerically adjacent integers can differ in many bits, causing large jumps and disrupting convergence on smooth problems.
- **Real-coded**: natural for continuous variables; no Hamming cliff; precision is not limited by bit length; better suited to smooth continuous optimisation.

Q1 (2025 variant). Additional Gradient-Free Questions

(h) [3 marks] The particle swarm update is:

$$x_{k+1}^{(i)} = x_k^{(i)} + v_{k+1}^{(i)}\Delta t, \qquad v_{k+1}^{(i)} = \alpha v_k^{(i)} + \beta\frac{x_{\text{best}}^{(i)} - x_k^{(i)}}{\Delta t} + \gamma\frac{x_{\text{best}} - x_k^{(i)}}{\Delta t}.$$

Define all symbols.

Solution:

- $x_k^{(i)}$: position of particle i at iteration k
- $v_k^{(i)}$: velocity of particle i at iteration k
- $x_{\text{best}}^{(i)}$: best position ever visited by particle i (personal best)
- x_{best}: best position ever found by *any* particle in the swarm (global best)
- α: inertia weight (momentum of previous velocity)
- β: cognitive coefficient (attraction to personal best)
- γ: social coefficient (attraction to global best)
- Δt: time step; k = iteration index; i = particle index

Q2. Gradient-Free Optimisation (Part II)

(a) [5 marks] Convert to standard LP form (minimise, equality constraints, $x \geq 0$):

$$\max\ 2x_1 - x_2 + 8x_3 \quad \text{s.t.} \quad 7x_1 + 5x_2 - x_3 \leq 1, \quad x_1 + 6x_2 \geq 4, \quad x_1, x_2 \geq 0,\ x_3 \text{ free.}$$

Solution: Four steps:

1. **Objective**: negate $\Rightarrow$ min $-2x_1 + x_2 - 8x_3$
2. **Free variable**: $x_3 = x_3^+ - x_3^-,\ x_3^+, x_3^- \geq 0$
3. $\leq$ **constraint**: add slack $s_1 \geq 0$: $7x_1 + 5x_2 - x_3^+ + x_3^- + s_1 = 1$
4. $\geq$ **constraint**: subtract surplus $s_2 \geq 0$: $x_1 + 6x_2 - s_2 = 4$

Standard form:

$$\min\ -2x_1 + x_2 - 8x_3^+ + 8x_3^- \quad \text{s.t.} \quad \underbrace{\begin{bmatrix} 7 & 5 & -1 & 1 & 1 & 0 \\ 1 & 6 & 0 & 0 & 0 & -1 \end{bmatrix}}_{A} \begin{bmatrix} x_1 \\ x_2 \\ x_3^+ \\ x_3^- \\ s_1 \\ s_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \end{bmatrix}, \quad \text{all variables} \geq 0.$$

(b) [4 marks] Solve the following optimal assignment problem using the Hungarian algorithm (find the assignment with minimum cost):

	Task 1	Task 2	Task 3
Worker A	100	150	125
Worker B	150	175	135
Worker C	120	250	140

Solution: Step 1 — row reduction (subtract row minimum: A=100, B=135, C=120):

$$\begin{pmatrix} 0 & 50 & 25 \\ 15 & 40 & 0 \\ 0 & 130 & 20 \end{pmatrix}$$

Step 2 — column reduction (subtract column minimum: col1=0, col2=40, col3=0):

$$\begin{pmatrix} 0 & 10 & 25 \\ 15 & 0 & 0 \\ 0 & 90 & 20 \end{pmatrix}$$

Step 3 — cover zeros: zeros at $(A, T_1), (B, T_2), (B, T_3), (C, T_1)$. Minimum cover: col T_1 + row B = 2 lines $< n = 3$. Not optimal.

Step 4 — update: min uncovered element = min$\{10, 25, 90, 20\} = 10$. Subtract 10 from uncovered cells; add 10 to doubly-covered cell (B, T_1):

$$\begin{pmatrix} 0 & 0 & 15 \\ 25 & 0 & 0 \\ 0 & 80 & 10 \end{pmatrix}$$

Step 5: zeros at $(A, T_1), (A, T_2), (B, T_2), (B, T_3), (C, T_1)$. Cover with row A + row B + col T_1 = 3 lines $= n$. Optimal.

Step 6 — assignment (independent zeros): $B \to T_3$ (unique zero in T_3), $C \to T_1$, $A \to T_2$.

Total cost: $150 + 135 + 120 =$ **405**.

(c) [2 marks] State the purpose of the relaxation step in branch-and-bound **and** give an example.

Solution: Purpose: solve the LP relaxation (drop integrality constraints) to obtain a bound on the optimal integer solution. If the relaxation bound is worse than the current best integer solution, the entire branch is pruned.

Example: in a binary programme with $x_i \in \{0, 1\}$, relax to $0 \le x_i \le 1$ (continuous LP). Solve; if $x_i^* \notin \{0, 1\}$, branch by adding $x_i \le 0$ and $x_i \ge 1$ as two sub-problems.

(d) [3 marks] Define the Pareto front and explain how it provides insights into an optimisation problem.

Solution: A solution x^* is **Pareto-optimal** if no other feasible solution simultaneously improves one objective without worsening at least one other. The **Pareto front** is the set of all such non-dominated solutions.

Insights: the front reveals the trade-off structure between competing objectives. A decision-maker can inspect the shape of the front to quantify trade-offs (e.g. how much cost increases per unit reduction in emissions) and select a preferred solution. A single-objective optimum lies at one extreme of the front.

(e) [6 marks] Show that maximising the log-likelihood is equivalent to minimising the sum of squared errors when measurement errors are i.i.d. $\mathcal{N}(0, \sigma^2)$.

Solution: Let $z_n = g_n - m(c, y_n)$ be the error at measurement n, distributed as $z_n \sim \mathcal{N}(0, \sigma^2)$. The likelihood is:

$$L(c) = \prod_{n=1}^{N} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{z_n^2}{2\sigma^2}\right).$$

Taking the log:

$$\ln L(c) = -\frac{N}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{n=1}^{N} z_n^2.$$

The first term is a constant (independent of c); $\frac{1}{2\sigma^2} > 0$ is a positive constant. Therefore maximising $\ln L$ over c is equivalent to minimising $\sum_{n=1}^{N} z_n^2$. ■

(2025 variant: simulated annealing) [3 marks] Briefly explain simulated annealing, incorporating the terms *acceptance probability* and *cooling schedule*.

Solution: Simulated annealing is a stochastic local search. At each iteration a candidate x' is generated near the current point x:

- If $f(x') < f(x)$: accept unconditionally.
- If $f(x') \geq f(x)$: accept with **acceptance probability** $P = \exp\big(-(f(x') - f(x))/T\big)$, where T is the current temperature.

The **cooling schedule** reduces T over time (e.g. $T_k = T_0 \cdot r^k$ for $r < 1$). High T allows uphill moves (exploration); as $T \to 0$ the algorithm converges to pure descent (exploitation).

Q3. Unconstrained Optimisation

(a)(i) [2 marks] Represent $i = 1, 2, 3, 4$ in base $b = 3$.

Solution: Using $i = a_0 + a_1 \cdot 3 + a_2 \cdot 9 + \cdots$:

$$1 = (1)_3, \quad 2 = (2)_3, \quad 3 = (10)_3, \quad 4 = (11)_3.$$

(a)(ii) [2 marks] Compute $\phi_i(3)$ for $i = 1, 2, 3, 4$ using $\phi_i(b) = \frac{a_0}{b} + \frac{a_1}{b^2} + \cdots$

Solution:

$$\phi_1(3) = \tfrac{1}{3}, \quad \phi_2(3) = \tfrac{2}{3}, \quad \phi_3(3) = \tfrac{0}{3} + \tfrac{1}{9} = \tfrac{1}{9}, \quad \phi_4(3) = \tfrac{1}{3} + \tfrac{1}{9} = \tfrac{4}{9}.$$

(b)(i) [3 marks] Using the second-order Taylor expansion of $f(x)$ around x_{k-1}, find the Newton update x_k.

Solution: Minimise the approximation $q(x)$ by differentiating and setting to zero:

$$\frac{\partial q}{\partial x} = \nabla f(x_{k-1}) + H(x_{k-1})(x - x_{k-1}) = 0 \;\Rightarrow\; \boxed{x_k = x_{k-1} - H(x_{k-1})^{-1} \nabla f(x_{k-1})}.$$

(b)(ii) [4 marks] Show that $(x_k - x_{k-1})$ lies in the direction of decreasing f when $H(x_{k-1})$ is positive definite.

Solution: Let $d = x_k - x_{k-1} = -H^{-1}\nabla f$. A direction is a descent direction if $\nabla f^\top d < 0$:

$$\nabla f^\top d = \nabla f^\top(-H^{-1}\nabla f) = -\nabla f^\top H^{-1} \nabla f.$$

Since H is positive definite (PD), H^{-1} is also PD, so $\nabla f^\top H^{-1} \nabla f > 0$ for $\nabla f \neq 0$. Therefore $\nabla f^\top d < 0$, confirming d is a descent direction. ■

(b)(iii) [4 marks] Using the quasi-Newton approximation $f(x) \approx f(x_k) + \nabla f(x_k)^\top (x - x_k) + \frac{1}{2}(x - x_k)^\top \tilde{H}(x_k)(x - x_k)$, write out the secant condition.

Solution: Differentiate the approximation with respect to x:

$$\nabla f(x) \approx \nabla f(x_k) + \tilde{H}(x_k)(x - x_k).$$

Set $x = x_{k-1}$:

$$\nabla f(x_{k-1}) = \nabla f(x_k) + \tilde{H}(x_k)(x_{k-1} - x_k).$$

Define $s_{k-1} = x_k - x_{k-1}$ and $y_{k-1} = \nabla f(x_k) - \nabla f(x_{k-1})$. Rearranging:

$$\boxed{\tilde{H}(x_k)\, s_{k-1} = y_{k-1}.}$$

This is the secant condition: the approximate Hessian must reproduce the observed change in gradient over the last step.

(c) [5 marks] Find x^* minimising the ridge regression objective $f(x) = \frac{1}{2}(y - Ax)^\top (y - Ax) + \frac{\lambda}{2} x^\top x$, $\lambda > 0$.

Solution: Differentiate and set to zero:

$$\nabla_x f = -A^\top (y - Ax) + \lambda x = A^\top A x - A^\top y + \lambda x = 0.$$

$$(A^\top A + \lambda I)\, x = A^\top y \;\Rightarrow\; \boxed{x^* = (A^\top A + \lambda I)^{-1} A^\top y.}$$

Since $\lambda > 0$, $(A^\top A + \lambda I)$ is positive definite, hence always invertible, giving a unique global minimiser.

2025 Q3b variant — the objective is: $L(x) = f(x_0) + \nabla f(x_0)^\top (x - x_0) + \frac{1}{2\mu}(x - x_0)^\top (x - x_0)$

(i) Gradient and Hessian:

$$\nabla_x L = \nabla f(x_0) + \tfrac{1}{\mu}(x - x_0), \qquad H_L = \tfrac{1}{\mu} I.$$

(ii) Set $\nabla_x L = 0$: $x^* = x_0 - \mu \nabla f(x_0)$ (gradient descent step with step size μ). Local minimum since $H_L = \frac{1}{\mu} I \succ 0$ for $\mu > 0$.

(iii) If $\mu < 0$: $H_L = \frac{1}{\mu} I \prec 0$ (negative definite) — L is a concave bowl opening downward with no lower bound. $\|x\|_2 \to \infty$; no bounded solution exists.

Q4. Power Systems Optimisation

(a) [4 marks] Estimate the time to solve AC OPF given: DC OPF solves in 5 s; 20 generators, 1000 buses, 1000 lines; $O((n + n_g + n_h)^3)$ scaling; nonlinearity gives a $5\times$ slowdown.

Solution: DC OPF: $\sim n = 1000$ bus angles, $n_h \sim 1000$ (power balance), $n_g \sim 1000$ (line limits). Complexity $\propto (n + n_g + n_h)^3 = 2000^3$, time = 5 s.

AC OPF: add voltage magnitudes and reactive power — roughly doubles variables and constraints $\Rightarrow n + n_g + n_h \approx 4000$:

$$t_{\text{AC}} = 5\,\text{s} \times \frac{4000^3}{2000^3} \times 5 = 5 \times 8 \times 5 = 200\,\text{s}.$$

(b) [6 marks] Suggest an algorithm for AC OPF and explain its main concepts.

Solution: Algorithm: Interior Point Method (IPM).

Justification: AC OPF is a large-scale non-convex NLP with both equality and inequality constraints. IPMs are efficient for large sparse structured problems (e.g. IPOPT used in industry).

Key concepts: convert inequality constraints $g(x) \le 0$ to equality using slack variables and add a logarithmic barrier:

$$\min_{x,s}\; f(x) - \mu \sum_i \ln s_i \quad \text{s.t.} \quad h(x) = 0,\; g(x) + s = 0,\; s > 0.$$

As $\mu \to 0$ the barrier solution converges to the constrained optimum. Each iteration solves a linear KKT system (Newton step on the first-order conditions):

$$\begin{bmatrix} H_{\mathcal{L}} & J_h^\top & J_g^\top \\ J_h & 0 & 0 \\ J_g & 0 & -S^{-1}Z \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta \nu \\ \Delta \lambda \end{bmatrix} = -r,$$

exploiting sparsity of the power network for efficiency.

(c) [5 marks] What is the difference between SC-OPF and regular AC OPF? Why is SC-OPF hard to solve in practice?

Solution: Regular AC OPF: minimise generation cost subject to power flow equations, generator limits, and thermal line limits for the base (intact) operating case only.

SC-OPF (N-1 security constrained): additionally requires the system to remain feasible after any single component outage (any one line or generator failure). Adds $N_{\text{components}}$ full sets of contingency constraints.

Why hard: for a 1000-bus, 1000-line system this means $\sim 10^3$ additional complete AC OPF problems. Direct formulation has $O(N_{\text{lines}})$ times more constraints, making it computationally intractable. Non-convexity also means each sub-problem may not solve to global optimality.

(d) [2 marks] How are security constraints included in industry practice without incurring huge computational costs?

Solution: Contingency screening: quickly rank all N contingency cases using a fast DC approximation. Select only the small subset of *binding* (worst-case) contingencies, then resolve AC OPF with those constraints added. Repeat iteratively until no new binding contingencies are found. This avoids solving the full SC-OPF directly.

(e) [8 marks] Formulate a non-linear least squares problem to estimate power system states from measurements $P_{12}, Q_{12}, P_{13}, Q_{13}, P_{32}, Q_{32}, V_1, V_2$, where:

$$P_{ij} = V_i V_j B_{ij} \sin(\theta_i - \theta_j), \qquad Q_{ij} = V_i (V_i - V_j \cos(\theta_i - \theta_j)) B_{ij}.$$

Solution: States: set $\theta_1 = 0$ as reference. Unknown vector: $x = [\theta_2,\, \theta_3,\, V_1,\, V_2,\, V_3]^\top$ (5 unknowns).
Measurement model $h(x)$:

$$h(x) = \begin{bmatrix} V_1 V_2 B_{12} \sin(\theta_1 - \theta_2) \\ V_1(V_1 - V_2 \cos(\theta_1 - \theta_2))B_{12} \\ V_1 V_3 B_{13} \sin(\theta_1 - \theta_3) \\ V_1(V_1 - V_3 \cos(\theta_1 - \theta_3))B_{13} \\ V_3 V_2 B_{32} \sin(\theta_3 - \theta_2) \\ V_3(V_3 - V_2 \cos(\theta_3 - \theta_2))B_{32} \\ V_1 \\ V_2 \end{bmatrix}, \quad z = \begin{bmatrix} P_{12} \\ Q_{12} \\ P_{13} \\ Q_{13} \\ P_{32} \\ Q_{32} \\ V_1 \\ V_2 \end{bmatrix}.$$

Optimisation problem (all meters equal variance σ^2):

$$\min_x \|z - h(x)\|_2^2 = \min_x \sum_{m=1}^{8} (z_m - h_m(x))^2.$$

Solved numerically using Gauss-Newton or Levenberg-Marquardt iteration.